

# Difference-in-Differences with Multiple Time Periods\*

Brantly Callaway<sup>†</sup>

Pedro H. C. Sant'Anna<sup>‡</sup>

December 1, 2020

## Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

**JEL:** C14, C21, C23, J23, J38.

**Keywords:** Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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\*First complete version: March 23, 2018. A previous version of this paper has been circulated with the title “Difference-in-Differences with Multiple Time Periods and an Application on the Minimum Wage and Employment”. We thank the Editor, the Associate Editor, two anonymous referees, Stéphane Bonhomme, Carol Caetano, Sebastian Calonico, Xiaohong Chen, Clément de Chaisemartin, Xavier D’Haultfoeuille, Bruno Ferman, John Gardner, Andrew Goodman-Bacon, Federico Gutierrez, Sukjin Han, Hugo Jales, Andrew Johnston, Vishal Kamat, Qi Li, Tong Li, Jason Lindo, Catherine Maclean, Matt Masten, Magne Mogstad, Tom Mroz, Aureo de Paula, Jonathan Roth, Donald Rubin, Bernhard Schmidpeter, Yuya Sasaki, Na’Ama Shenhav, Tymon Słoczyński, Sebastian Tello-Trillo, Alex Torgovitsky, Jeffrey Wooldridge, Haiqing Xu and several seminar and conference audiences for comments and suggestions. Code to implement the methods proposed in the paper is available in the R package `did` which is available on CRAN.

<sup>†</sup>Department of Economics, University of Georgia. Email: brantly.callaway@uga.edu

<sup>‡</sup>Department of Economics, Vanderbilt University. Email: pedro.h.santanna@vanderbilt.edu

# 1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).<sup>1</sup>

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group  $g$  at time  $t$ , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

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<sup>1</sup>See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

adoption in the literature. We interpret this assumption as if units do not “forget” about the treatment experience.<sup>4</sup>

Define  $G$  as the time period when a unit first becomes treated. Under Assumption 1, for all units that eventually participate in the treatment,  $G$  defines which “group” they belong to. If a unit does not participate in any time period, we arbitrarily set  $G = \infty$ . We define  $G_g$  to be a binary variable that is equal to one if a unit is first treated in period  $g$  (i.e.,  $G_{i,g} = \mathbf{1}\{G_i = g\}$ ) and define  $C$  to be a binary variable that is equal to one for units that do not participate in the treatment in any time period (i.e.,  $C_i = \mathbf{1}\{G_i = \infty\} = 1 - D_{i,\mathcal{T}}$ ). Let  $\bar{g} = \max_{i=1,\dots,n} G_i$  be the maximum  $G$  in the dataset. Next, denote the generalized propensity score as  $p_{g,s}(X) = P(G_g = 1 | X, G_g + (1 - D_s)(1 - G_g) = 1)$ . Note that  $p_{g,s}(X)$  indicates the probability of being first treated at time  $g$ , conditional on pre-treatment covariates  $X$  and on either being a member of group  $g$  (in this case,  $G_g = 1$ ) or a member of the “not-yet-treated” group by time  $s$  (in this case,  $(1 - D_s)(1 - G_g) = 1$ ). Many of our results use a specialized version of this generalized propensity score, and, henceforth, we define  $p_g(X) = p_{g,\mathcal{T}}(X) = P(G_g = 1 | X, G_g + C = 1)$  which is the probability of being first treated in period  $g$  conditional on covariates and either being a member of group  $g$  or not participating in the treatment in any time period. Let  $\mathcal{G} = \text{supp}(G) \setminus \{\bar{g}\} \subseteq \{2, 3, \dots, \mathcal{T}\}$  denote the support of  $G$  excluding  $\bar{g}$ .<sup>5</sup> Likewise, let  $\mathcal{X} = \text{supp}(X) \subseteq \mathbb{R}^k$  denote the support of the pre-treatment covariates. Finally, for a generic  $\delta \geq 0$ , let  $\mathcal{G}_\delta = \mathcal{G} \cap \{2 + \delta, 3 + \delta, \dots, \mathcal{T}\}$ .

Next, we set up the potential outcomes framework. Here, we combine the dynamic potential outcomes framework of Robins (1986, 1987) with the multi-stage treatment adoption setup discussed by Heckman et al. (2016); see also Sianesi (2004). Let  $Y_{i,t}(0)$  denote unit  $i$ ’s untreated potential outcome at time  $t$  if they remain untreated through time period  $\mathcal{T}$ ; i.e., if they were not to participate in the treatment across all available time periods. For  $g = 2, \dots, \mathcal{T}$ , let  $Y_{i,t}(g)$  denote the potential outcome that unit  $i$  would experience at time  $t$  if they were to first become treated in time period  $g$ . Note that our potential outcomes notation accounts for potential dynamic treatment selection, though it also accommodates (pre-specified) treatment regimes (Murphy et al., 2001; Murphy, 2003). The observed and potential outcomes for each unit  $i$  are related through

$$Y_{i,t} = Y_{i,t}(0) + \sum_{g=2}^{\mathcal{T}} (Y_{i,t}(g) - Y_{i,t}(0)) \cdot G_{i,g} \quad (2.1)$$

In other words, we only observe one potential outcome path for each unit. For those that do not participate in the treatment in any time period, observed outcomes are untreated potential outcomes in all periods. For units that do participate in the treatment, observed outcomes are the unit-specific potential outcomes corresponding to the particular time period when that unit adopts the treatment.

We also impose the following random sampling assumption.

**Assumption 2** (Random Sampling).  $\{Y_{i,1}, Y_{i,2}, \dots, Y_{i,\mathcal{T}}, X_i, D_{i,1}, D_{i,2}, \dots, D_{i,\mathcal{T}}\}_{i=1}^n$  is independent and identically distributed (*iid*).

Assumption 2 implies that we have access to panel data; our results extend essentially immediately

<sup>4</sup>See Han (2020), de Chaisemartin and D’Haultfoeulle (2020) and Bojinov et al. (2020) for alternative setups where treatment can “turn off”.

<sup>5</sup>When there is a “never treated” set of units with  $G = \infty$ ,  $\mathcal{G}$  only excludes this group. When such “never-treated” group is not available, we exclude the latest-treated group as there will be no valid untreated comparison group for them.